

IMPLICATIONS OF TECHNOLOGICAL AND POLICY CHANGES ON USE OF NITROGENOUS FERTILIZERS IN INDIA

A project report submitted by

ANMOL MAHAJAN (PGDM-ABM-2106)

In partial fulfillment for the award of

Post Graduate Diploma in Management (Agribusiness Management)

under the supervision of

DR. RANJIT KUMAR



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ICAR-NATIONAL ACADEMY OF AGRICULTURAL RESEARCH MANAGEMENT
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APRIL 2023

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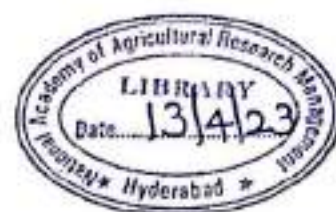
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BONAFIDE CERTIFICATE

Certified that this project report **"IMPLICATIONS OF TECHNOLOGICAL AND POLICY CHANGES ON USE OF NITROGENOUS FERTILIZERS IN INDIA"** is the bonafide work of **"ANMOL MAHAJAN"** who carried out the project work under my supervision.

SIGNATURE

Dr. RANJIT KUMAR

SUPERVISOR

Principal Scientist

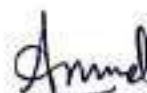
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STUDENT DECLARATION

Date: 10-04-2023

This is to declare that I, **Anmol Mahajan**, a student of Post Graduate Diploma Programme in Management (Agri-Business Management) (2021-2023), ICAR-NAARM have given original data, information and maintained full confidentiality to the best of my knowledge in the project report titled "**Implications of Technological and Policy Changes on Use of Nitrogenous Fertilizers in India**" and that, no part of this information has been used for any other assignment but for the partial fulfilment of the requirements towards the completion of the said project.



ANMOL MAHAJAN

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ICAR-NAARM

EXECUTIVE SUMMARY

ACKNOWLEDGEMENT

This acknowledgement is not merely a catalogue of names but an expression of a deep sense of gratitude to all those who helped me in this project directly or indirectly.

I take this opportunity to express my sincere and deepest gratitude to **ICAR-NAARM, HYDERABAD** for laying the building blocks of logic and pragmatism in my life. This report, in a way reflects these values. I would like to express my deepest gratitude and reverence to **Dr. Ranjit Kumar (Principal Scientist, ABM Division, ICAR-NAARM)** for his excellent leadership, original and compassionate direction of the difficult task, and love and care he displayed throughout the course of the project.

Last but not the least; I bow my head in front of the almighty and my parents who has always been the torch bearer for me throughout my work, by showing me the right path and boosting my moral to bring the best out of me.

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EXECUTIVE SUMMARY

Out of 17 nutrients, which are essentially required by crop plants for their normal growth and reproduction, Nitrogen (N) is generally required in largest quantum. Nitrogen (N) is a key input to food production. The use of nitrogenous fertilizers has been a critical aspect of modern agriculture, particularly in developing countries like India. Nearly half of N fertilizer input is not used by crops and is lost into the environment via emission of gases or by polluting water bodies. Excess usage of nitrogenous fertilizers has led to widening of NPK ratio and adversely affected the soil health. Urea is one of the most widely used sources of fertilizer N in the world. However, the rapid pace of technological and policy changes in recent years has led to significant shifts in the way these fertilizers are produced and utilized. This study aims to examine the implications of these changes on the use of nitrogenous fertilizers in India, with a focus on the impact of technology advancements, policy changes, and their interplay on the Indian agricultural sector. The study will consider the implications of recent policy changes on nitrogen fertilizer use, as well as the role of government and other stakeholders in promoting balanced use of fertilizers. The findings of this research will provide valuable insights into the challenges and opportunities facing India's fertilizer sector and inform policy decisions aimed at promoting sustainable agriculture and food security in the country.

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IFCO – Indian Farmers Fertiliser Cooperative Limited

GCA – Gross-Cropped Area

BCR – Benefit-Cost Ratio

ROI – Return on Investment

LIST OF ACRONYMS AND ABBREVIATIONS

MT - Metric tonnes

Ha - Hectare

Mn - Million

FAO - Food and Agriculture Organisation of the United Nations

FAI - Fertilizer Association of India

IFA - International Fertilizer Association

NOCU - Neem Oil Coated Urea

FUE - Fertilizer Use Efficiency

NUE - Nutrient Use Efficiency

NBS - Nutrient Based Subsidy

RPS - Retention Price Scheme

DAP - Diammonium Phosphate

MOP - Muriate of Potash

NPK - Nitrogen, Phosphorus, and Potassium

IFFCO - Indian Farmers Fertiliser Cooperative Limited

GCA - Gross- Cropped Area

BCR - Benefit-Cost Ratio

ROI - Return on Investment

CHAPTER 1 – INTRODUCTION

1.1 Introduction

Agriculture contributes about 18.8 per cent to the Gross Domestic Product (GDP) and provides a survival to near about two third of the Indian Population. With the growing population, it is important for the economy to secure its population with food security. From traversing food deficit and the food grain import country during and after independence, India had slowly reshaped to food surplus, self-sufficient and net exporter of food grains in the global scenario. The advent of green revolution led a stone of the use of technology, high yielding variety seeds, synthetic fertilizer and irrigations to improve production and productivity. The food grain production increased from 52 million tons in 1951-52 to 315.7 million tons in 2021-22.

Fertilizer production and consumption has been increasing in India over the decades and currently it is the second largest consumer of complex synthetic fertilizer i.e. NPK after China and third largest producer after China and USA. As the seventh largest nation in the world with 329.7 million hectare area, it has a net cultivable area of 141 million hectares and cropping intensity of 135 per cent. Fertilizer nitrogen (N) has contributed an estimated 40% to the increase in per capita food production over the past 50 years (Brown, 1999; Smile, 2002). Nitrogen (N) fertiliser demand increased dramatically from 0.14 thousand tonnes in 1970–1971 to 17.63 million MT in 2018–19. (FAI 2020). Nitrogen fertilizer is tamed to be both essential nutrients in increasing crop yield and quality and pollutant in terrestrial ecosystem (Khajuria 2016).

Fertilizer consumption of India has increased from 16.6 kg/ha in 1971 to 209.4 kg/ha in 2020 growing at an average annual rate of 5.70% (FAI). In the financial year 2021-22, nitrogenous fertilizers accounted for over 62.71% of the total fertilizer use in India (Statista, 2021). There has been commensurate increase in production, import and consumption of urea over the years. With consumption of fertilizer-nitrogen (N) increasing from 0.6 Million tonnes (Mt) in 1965-66 to 19.44 Mt in 2020-21, India has emerged as the second largest consumer of N in the world. Since 1970s, urea is the

major source of fertilizer-N; 79 % fertilizer-N was consumed as urea in 2020-21 (FAI, 2021). Its supremacy as India's primary source of fertiliser N is projected to persist thanks to record-breaking urea output of 25.08 Mt in 2021-22. While urea provides the most N at the lowest Urea cost, has no storage risks and can be used for all types of crops and soils with little or no harm to the soil, use efficiency of urea-N by different crops can be as low as 20% and it rarely exceeds 50%.

There has been inter- state, -district, -crop -area and – village variation in fertilizer consumption inclusive of urea consumption. Balanced use of fertilizers is key to enhance farm productivity and for sustainable soil health. Fertilizer consumption in India is imbalanced and skewed in favour of Urea-N. All India NPK consumption ratio has widened from 4:3.2:1 in 2009-10 to 7.7:3.1:1 in 2021-22 (FAI, 2021, 2022). The NPK ratio in the Indo-Gangetic region is abruptly low with response to the huge food grain contribution. In Punjab it is 61.7:19.2:1, in Haryana 61.4:18.7:1, in Uttar Pradesh 25.2:8.8:1. The present imbalances in the use of NPK ratio have created the problem in declining soil productivity and farmers Income, which are both economic and environmental losses.

Following the Green Revolution, fertiliser policies were created with the dual goals of increasing domestic fertiliser production capacity and providing fertilisers to farmers at reasonable costs. During this period, state policy ensured an increase in the supply of raw materials required for manufacturing fertilizers. The public sector and cooperatives led the production of fertilizers, and both fertilizer prices and the distribution of fertilizers across states were regulated by the government. The system of controls and subsidies was designed to provide assured returns to fertilizer manufacturers as well as keep fertilizer prices low for farmers. Significant changes were brought about in these policies after the 1990s. These have had profound implications for Indian agriculture. Decontrolling the prices of all non-urea fertilizers was the most significant change brought about during the post-liberalisation period. This resulted in a surge in the prices of fertilizers and a bias towards urea consumption.

The Nutrient Based Subsidy plan was a change in the subsidy system announced by the Indian government in 2010 as part of its ongoing attempts to decontrol fertiliser prices.

By severing the link between the subsidy regime and the pricing of chemical fertilisers, this method allowed for the total deregulation of non-urea fertiliser prices. The Indian peasantry will suffer greatly as a result of this policy move, which has caused a sharp increase in the cost of non-urea fertilisers.

1.2 Need for study

A rise in fertilizer consumption over the years, along with the adoption of HYVs and the expansion of groundwater irrigation facilities, has resulted in highest food grain production in India. Nitrogenous fertilizers have played a crucial role in improving agricultural productivity and ensuring food security of the country. However, excessive and imbalanced use of nitrogenous fertilizers has led to environmental degradation such as multi-nutrient deficiencies and deterioration of soil health, and thereby deficiency symptoms in plants. Further, there exists a huge variation in fertilizer application across various states and regions in India. This study analyses the impact of recent policy changes on fertilizer consumption pattern in India.

1.3 Objectives

Following are the objectives of the study:

1. To examine the use pattern of different fertilizers in India.
2. To analyse the impact of recent policy changes like introduction of neem coated urea (NCU) and reducing the weight of bags of urea on fertilizer consumption.
3. To analyse the adoption behaviour of Nano urea among selected farmers.

CHAPTER 2- REVIEW OF LITERATURE

Literature studies aid in understanding the present situation and its difficulties as well as in locating deep issues. It provides an instant overview of the pertinent research that have been conducted in the past while keeping the study's purpose in mind.

- **Desai (1986)** through a policy study for growth of consumption, opined that constraint in the lowering real price of fertilizer will determine the consumption. Further, he added that non-price factors such as improving the efficiency, shifting the response by improving the deficiency of supply and distribution system, liberalizing the policy environment for agriculture by subsidizing the nitrogen-based fertilizer at a higher rate has driven the consumption.
- **Shivay et al. (2001)** from ICAR - Indian Agricultural Research Institute, New Delhi, based on two years study reported 7.81% increase in the rice grain yield with application of 0.5% of 10% neem emulsion coated urea compared to prilled urea. Further, the nitrogen recovery efficiency with application of same treatment was 39.1% compared to 26.9% with prilled urea alone.
- **Tiwari (2001)** addressed the issue of reduction in yields due to overuse of urea. He shows that using just urea, and not other fertilizers, leads to declines in yields over time as naturally occurring nutrients in the soil are exhausted. He pushes for policies that recognize this problem and incentivize balanced fertilizer usage. The Nation Academy of Agricultural Studies (NAAS 2009) paper on crop response and nutrient ratio goes even further. It argues that the classical 4:2:1 ratio for application of N, P, and K may no longer be valid. The paper suggests that even higher levels of P and K may be necessary in some regions.
- **Fan et al. (2007)** lay the main arguments against input subsidies. They highlight the critical role it played during the Green Revolution but conclude that "Agricultural input and output subsidies have proved to be unproductive, financially unsustainable, environmentally unfriendly in recent years, and contributed to increased inequality among rural Indian states" (emphasis added).

They call for subsidies to be cut back while increasing spending on agricultural research and development, rural infrastructure, and education. While they make a technically sound argument, they do not consider how such a policy might be implemented given India's political economy.

- According to **Gu et al. (2011)**, technological developments play a key role in atmospheric N pollution control while effective policies mainly contribute to groundwater N pollution control. Therefore, the development and implementation of regulations is critical to orientate N fertilization practices that minimize the environmental impact. Although varying N management policies have been implemented for some time in different countries, they have had variable degrees of success in achieving required food productions levels while protecting the environment.
- **Singh (2016)** did a detailed review of studies on the efficacy of neem oil coated urea for improving yields and increasing nitrogen use efficiency. He showed that for the fifty-five studies on paddy, the average increase in yields because of neem oil coating was 6.3 per cent. For wheat, the author reviewed seven studies that found an average yield increase of 5.3 per cent as a result of using neem oil coated urea. In about thirty per cent of the studies for paddy and wheat, neem oil coated urea did not outperform uncoated urea. The average percentage increase in yield on account of neem oil coated urea was 10.5 per cent in potato, 8.7 per cent in sugarcane, 4.3 per cent in cotton and 5.4 per cent in finger millet. The studies on maize found no increase at all.
- **Gulati et al. (2015, 2018)** criticized the subsidy policy adopted in the fertilizer sector over the years by stating that it created a substantial financial burden to the Central Government and keeping the urea prices low; it also failed to encourage a balanced use of fertilizers in the soil.
- In a recent study carried out by **Ernst and Young (E&Y)** sponsored by the Department of Fertilisers, Government of India, on the NBS policy, it was recommended that "reasonability" of MRP for P & K fertilisers must be done considering two criteria, namely, "profitability and return for the capital employed for the industry players" and "spend on fertilisers for key crops as a %

of farmer's income". The report also suggested bringing urea in the NBS regime and "strategic investments by Indian players in mines abroad by creating a 'sovereign fund'" for P&K fertilisers to ensure their supply in the country.

- **Kumar et al., (2020)** confirms from results of field trials that NUE can be improved by synchronizing the nutrient availability as per the need of the crops. Here, foliar application of Nano fertilizers at critical growth stages of crops can provide opportunity for improving nutrient utilisation, arresting/minimising nutrient losses from the crop production system.
- **Kumar et. al. (2021)** summarized the results on application of Nano urea – liquid (Nano N) across various locations. The results revealed that two foliar applications at critical growth stages of crops like rice, wheat, maize, tomato, cucumber and capsicum led to a reduction in the application rate of Fertilizer-N and also caused increase in the yield in range of: 3-23% in wheat, 5-11% in tomato, 3-24% in paddy/rice, 2-15% in maize, 5% in cucumber, and 18% in capsicum. Foliar application of Nano urea, thus, increased the number of effective tillers, enhanced the growth and raised the biomass yield (both grain and straw yields) in most of the crops at different locations. Economic benefits (BCR or ROI) of Nano fertilizer application have to be seen from the perspective of extra yield achieved along with reductions in fertilizer usage as well as direct and indirect benefits in terms of reduction in the environmental (soil-air-water) pollution.

CHAPTER 3 – DESIGN OF THE STUDY

3.1 Methodology

The present study is based on both primary and secondary research.

3.2 Collection of Data

Primary data pertaining to fertilizer consumption pattern, awareness about neem-coated Urea and benefits of Nano Urea was collected from farmers. Secondary data pertaining to the fertilizer consumption in the world was collected from the website of International Fertilizer Association and Statista. The data pertaining to fertilizer consumption pattern in India and crop-wise fertilizer consumption was collected from the website of Fertilizer Association of India and Input Survey report of 2016-17. The data pertaining to trend in fertilizer subsidy over the years and retail prices of fertilizers was collected from the annual reports of Fertilizer Statistics by Fertilizer Association of India (FAI).

3.3 Limitations of the study

Lack of awareness about the usage of Nano-urea is a major limitation in this study. It was observed from the investigation that majority of the farmers in the sample areas were not aware of the benefits of Nano urea. Besides, the study has got its own limitation in the sense that data was collected through telephonic surveys. So, personal connect with farmers was missing in this study. Also, some of the secondary data sources are not updated and therefore can only be used as reference. Since, the study concerns the fertilizer industry which can be affected a number of factors, the available data cannot be exactly used to reach a conclusion.

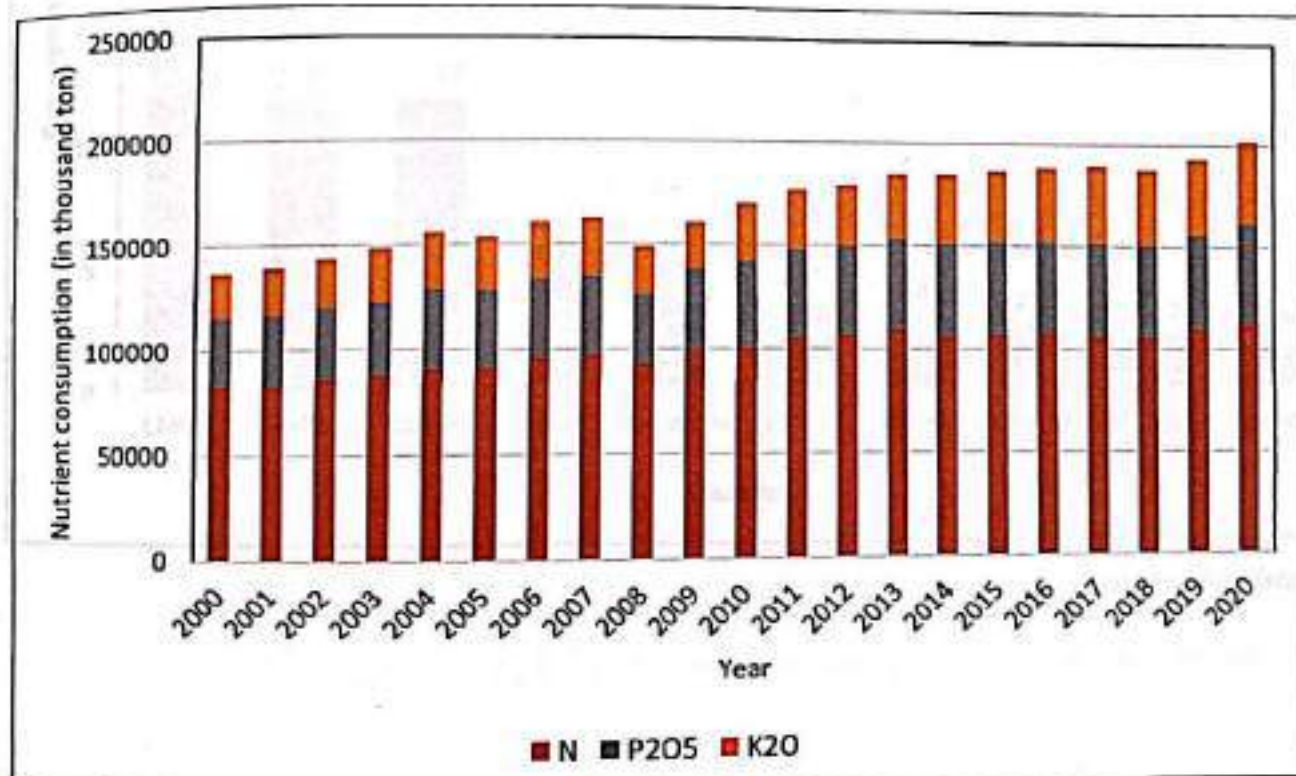
CHAPTER 4-RESULTS AND DISCUSSION

4.1 Trends in Fertilizer Consumption

4.1.1 Global Scenario

From (Figure 1), the global nutrient consumption has increased steadily between 2000 and 2020, by 47.38%, rising from 136.94 million tonnes to 201.83 million tonnes of nutrients ($N+P_2O_5+K_2O$). During the same period, global N nutrient consumption has increased by 36.92%, from 82.06 to 112.36 million tonnes of N, P consumption has increased by 49.25% and K consumption increased by 83.48% reporting the highest increase. The consumption of fertilizers has decreased only during 2008 in the last 20 years as that year was marked by global financial crisis.

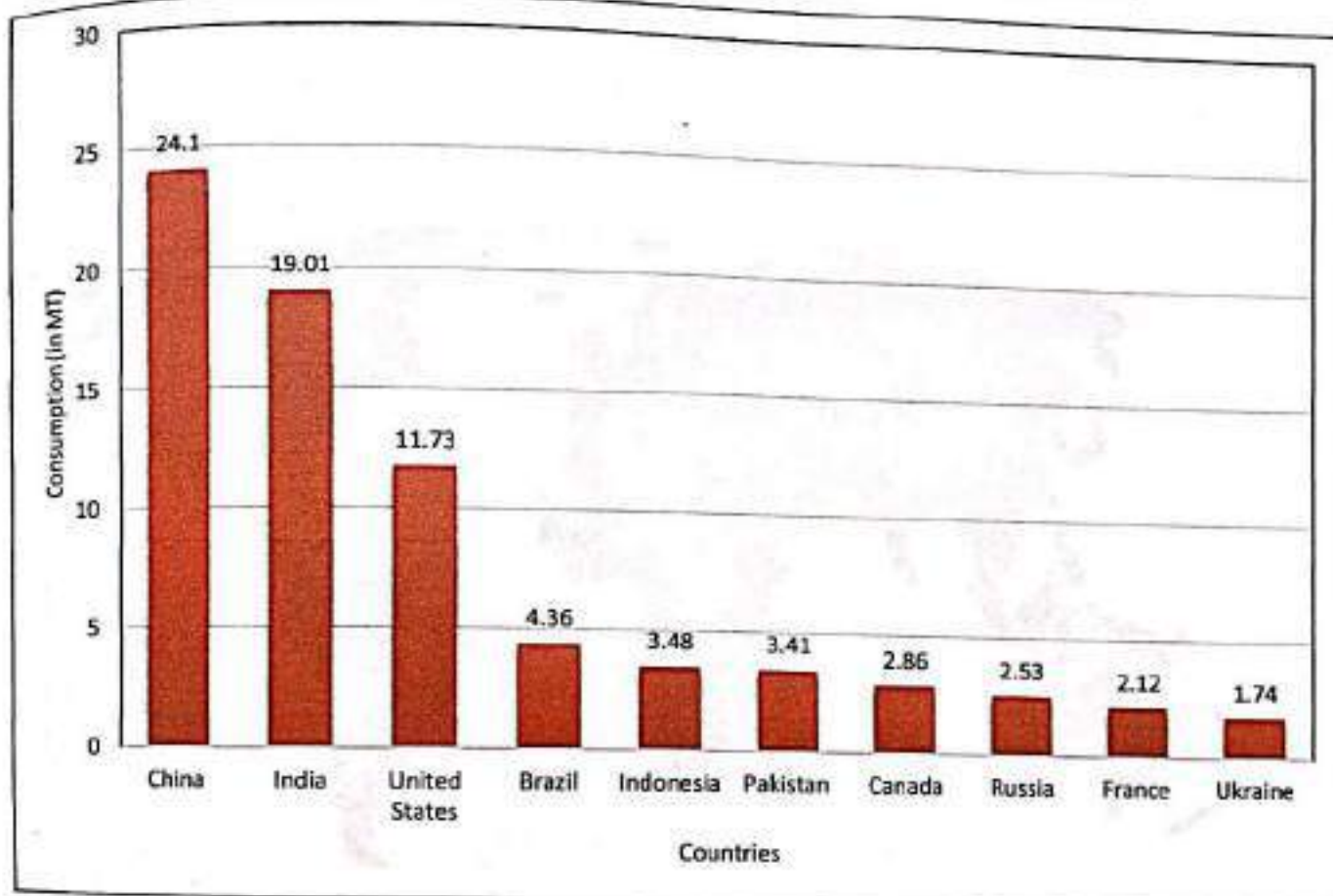
Figure 1 Global Fertilizer Consumption



Source: IFA

From (Figure 2), it was noted that China is the largest consumer of nitrogenous fertilizers in the world with a consumption of 24.1 million MT of nutrient N in 2019-20. The second leading country in nitrogenous fertilizer consumption is India by 19.01 million tonnes followed by United States (11.73 million tonnes), Brazil (4.36 million tonnes) and Indonesia (3.48 million tonnes).

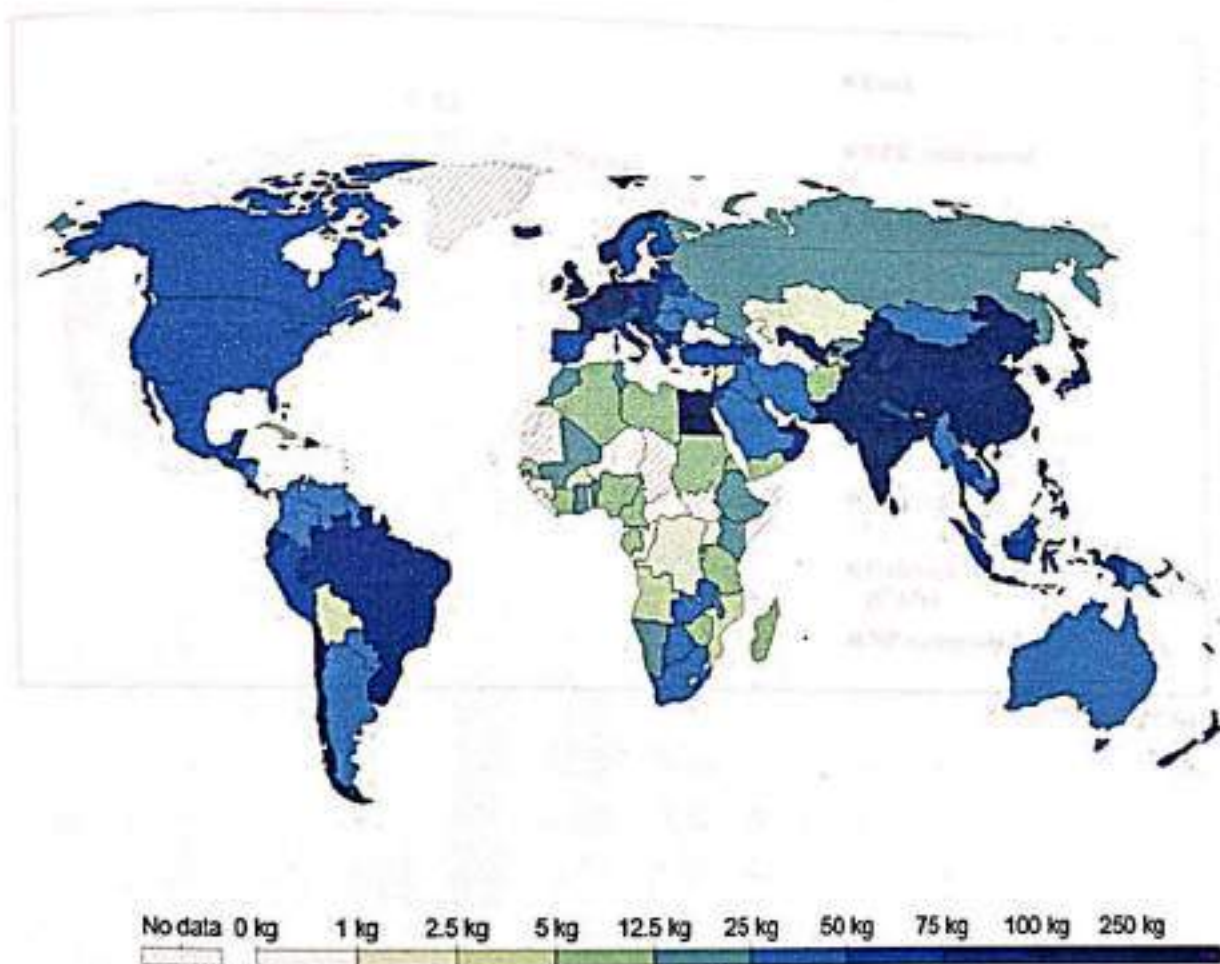
Figure 2 Top 10 consumers of nutrient N in the world in 2019-20



Source: Statista

From (Figure 3), East Asia had the largest per hectare consumption of nitrogenous fertilizers worldwide followed by South Asia and Latin America in 2019-20. South-East Asian countries like India, Pakistan and Bangladesh had more than 150 kg/ha nitrogenous fertilizer consumption which is highest in the world. Nitrogen fertilizer consumption per hectare was less than 50 kg/ha in most of the African countries.

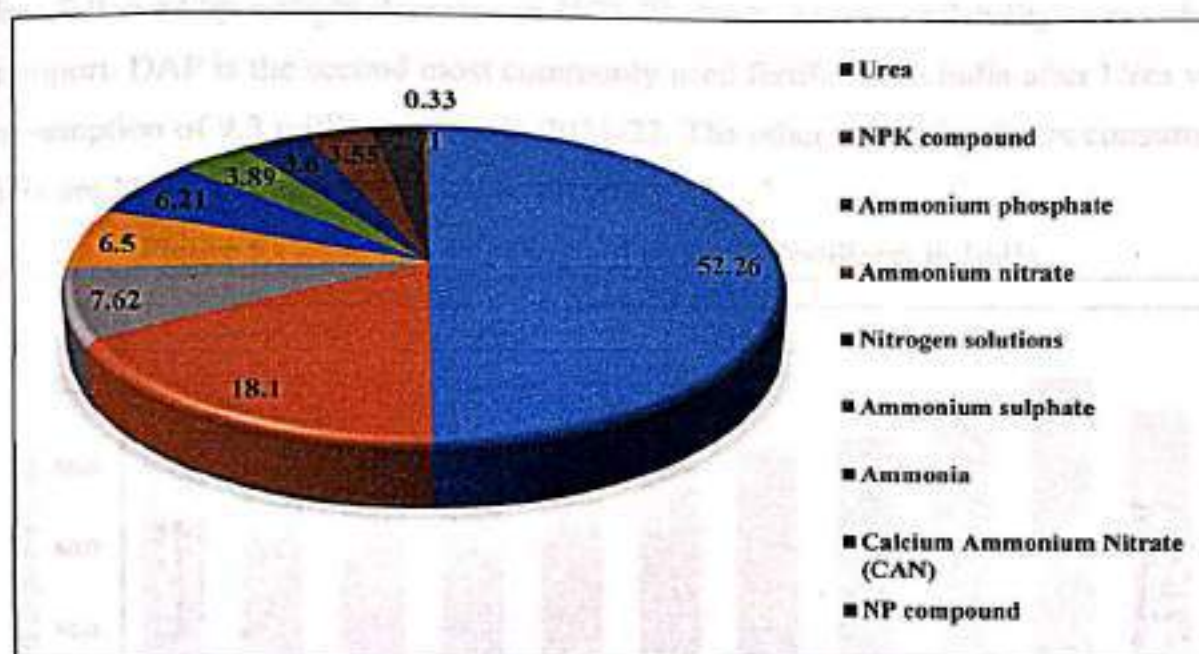
Figure 3 Nitrogen fertilizer consumption per hectare of cropland, 2019



Source: UN FAO

The global nitrogenous fertilizer market is largely dominated by urea, which accounted in 2019 for 52.26% of global N fertilizer use (Figure 4). The wide acceptance of urea is because of its agronomic acceptability and relatively of its lower cost as compared to other chemical fertilizers. It also has a high nitrogen content (46%), relative to many other popular nitrogen sources. The other important forms of nitrogen fertilizer produced and consumed in the world include Ammonium Sulphate (AS), Calcium Ammonium Nitrate (CAN) and Ammonium Chloride.

Figure 4 Global nitrogen fertilizer consumption by product (2019)

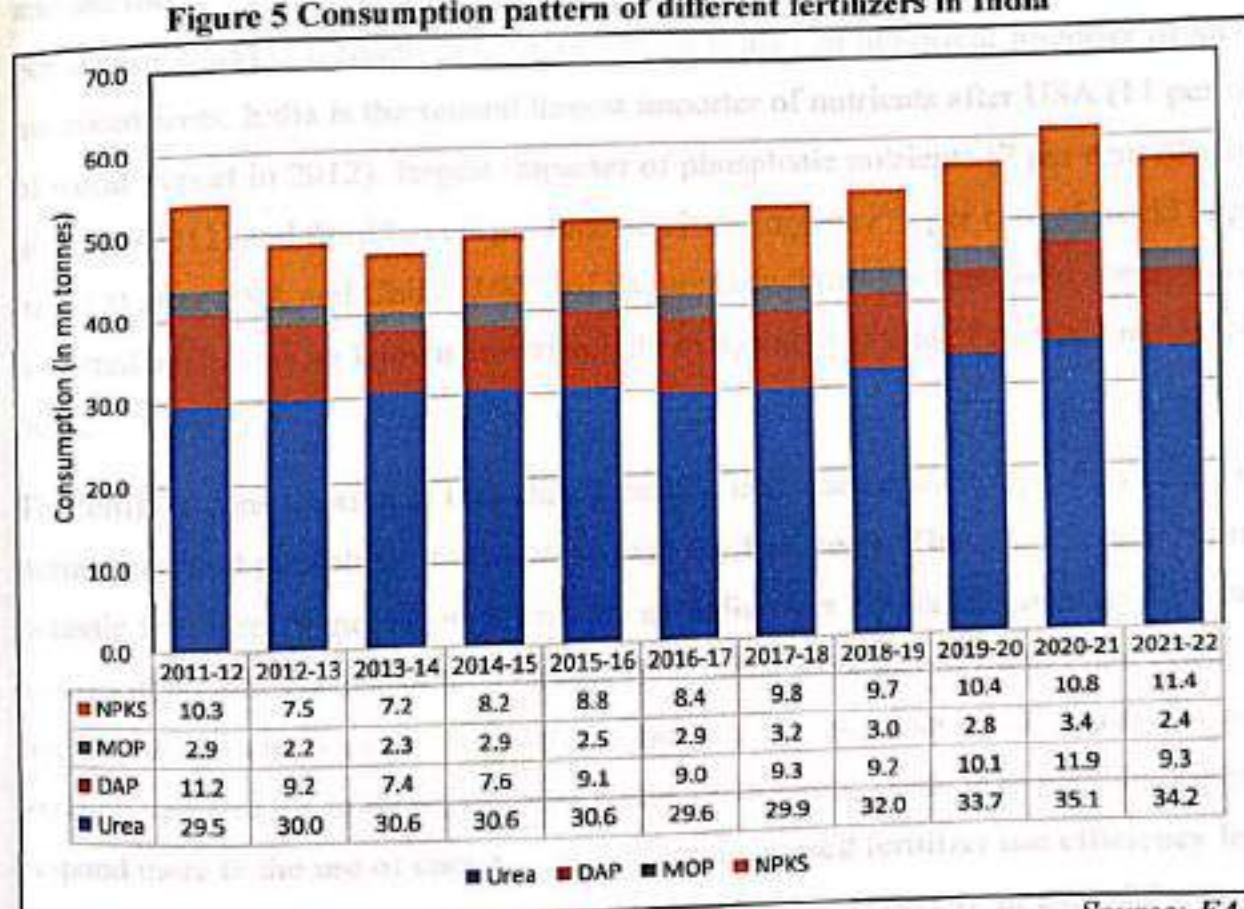


Source: Statista

4.1.2 Indian Scenario:

Urea is the most consumed chemical fertilizer in India, accounting for about 60% of the total chemical fertilizer consumption in the country. In India, the consumption of Urea has increased from 29.5 million tonnes in 2011-12 to 34.2 million tonnes in 2021-22 (Figure 5). The consumption of Urea in 2020-21 was recorded as highest i.e., 35.1 million tonnes. There was a steady increase in Urea consumption from 2011-12 to 2015-16. Urea consumption declined by 3.33 per cent from 30.6 million tonnes in 2015-16 to 29.6 million tonnes in 2016-17 due to introduction of 100% Neem Oil Coated Urea. Between 2016-17 and 2020-21, there has been a steady increase in the consumption of Urea followed by a slight decrease in 2021-22 due to lower availability caused by cut in import. DAP is the second most commonly used fertilizers in India after Urea with a consumption of 9.3 million tonnes in 2021-22. The other major fertilizers consumed in India are MOP and NPK complex fertilizers.

Figure 5 Consumption pattern of different fertilizers in India



Source: FAI

India is the second largest consumer of fertilizers in the world, after China. Fertilizer consumption in India has increased by about 13 times during from but fertilizer use has remained skewed in favour of N. With ushering into the Green Revolution era in mid-1960s through the introduction of fertilizer-responsive high-yielding varieties of rice and wheat, fertilizer consumption in India witnessed a spectacular rise.

India is an important player in the world market for fertilisers. For the last decade (2010-2019), India has been the second highest producer of nitrogenous fertilisers after China (producing 10-11 per cent of world production) and third highest producer of phosphatic fertilisers after China and USA (producing around 7 per cent of total world production in 2012). India is important as a consumer as well. It is the second biggest consumer of nitrogenous and phosphatic fertilisers after China (17.68 per cent and 16.59 per cent respectively of world consumption in 2019 for nitrogenous and phosphatic fertilisers) and the fourth biggest consumer of potassic fertilisers after China, Brazil and USA (7 per cent of world consumption in 2019). India is also an important importer of all the macronutrients. India is the second largest importer of nutrients after USA (11 per cent of world import in 2012), largest importer of phosphatic nutrients (7 per cent of world import in 2012) and third largest potassic nutrients importer (6 per cent of world import in 2012) after USA and China. 100% of the potassic fertilisers that India consumes are imported as there is no known commercially extractable source of potassic nutrients in India.

The fertiliser consumption in India has generally exceeded domestic production in both nitrogenous and phosphatic fertilisers except for few years. The entire requirement of potassic fertilisers is met through imports as India does not have commercially viable sources of potash. The use of fertilizers is affected by a number of factors like irrigation, high yielding variety seeds, size of the farm credit etc. Increased area under high yielding varieties led to increased food grains production. These high yielding varieties respond more to the use of chemical fertilizers. Increased fertilizer use efficiency leads to a number of benefits to Indian agriculture. They are economy in use of fertilizers, reduction in unit cost of production, prevention of fall in agricultural productivity,

production of environmental quality and efficient use of other inputs such as irrigation and high yielding varieties in developing countries actual fertilizer use is usually below the economic potential. The rapid expansion of irrigation, spread of HYV seeds, introduction of retention price scheme, distribution of fertilizers to farmers at affordable prices, expansion of dealer's network, improvement in fertilizer availability and virtually no much change in farm gate fertilizer prices for recent years were major reasons for increase in fertilizer consumption. The studies by Vijay and Hrima (2011), Jayanthi et al. (2013) also contributed the similar findings.

Figure 4 Consumption pattern of N per hectare of GCA in major states in 2015-16

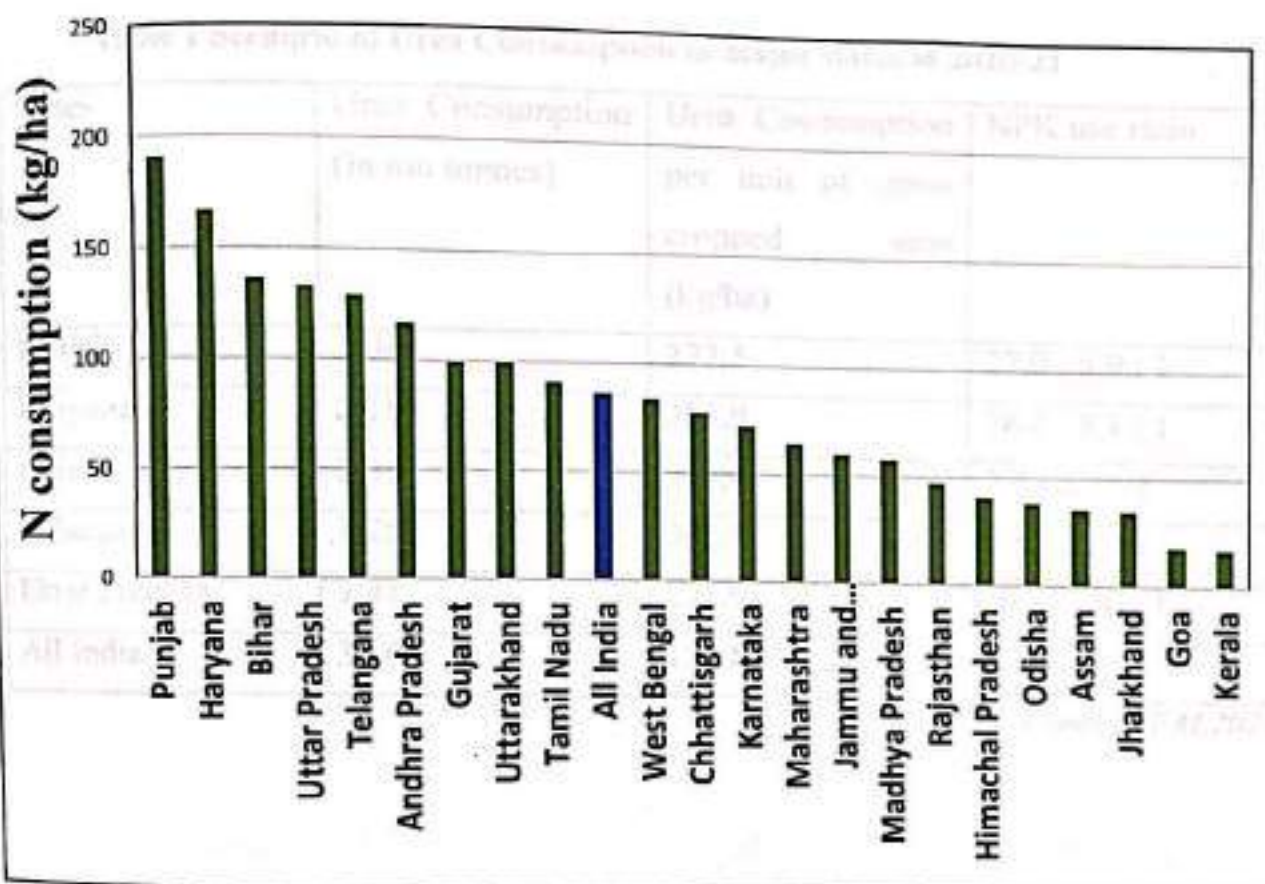


Source: Agricultural Statistics, 2017

4.1.3 Fertilizer Use in India Among States

The all India N consumption per hectare of GCA was 86.01 kg/ha in 2020-21. Punjab was the highest N consuming state per hectare of GCA with a consumption of 190.81 kg/ha followed by Haryana and Bihar with a consumption of 167.05 and 136.42 kg/ha respectively in 2020-21. Kerala was the lowest nutrient N consuming state with a consumption of 17.35 kg/ha in 2020-21, which was almost 10-fold less than Punjab. (Figure 6).

Figure 6 Consumption pattern of N per hectare of GCA in major states in 2020-21



Source: Agricultural Statistics, 2021

Urea consumption per hectare of gross cropped area in India was 173.8 kg/ha in 2020-21. From **Table 1**, the total consumption of Urea was found to be highest in Uttar Pradesh, that is 7.41 MT in 2020-21. The state-wise analysis of Urea consumption in India shows that green revolution states having predominant rice- wheat cropping system have more consumption of urea. Urea consumption per hectare of GCA was found to be highest in Punjab, that is 377.5 kg/ha followed by Haryana (325.9 kg/ha) and Bihar (314.6 kg/ha) in 2020-21. NPK ratio was also found highly skewed towards N in Punjab with 27.0 : 6.9 : 1 as against all India NPK ratio of 6.5 : 2.8 : 1 in 2020-21.

Table 1 Scenario of Urea Consumption in major states in 2020-21

States	Urea Consumption (in mn tonnes)	Urea Consumption per unit of gross cropped area (kg/ha)	NPK use ratio
Punjab	2.94	377.5	27.0 : 6.9 : 1
Haryana	2.15	325.9	28.2 : 8.1 : 1
Bihar	2.33	314.6	7.7 : 2.7 : 1
Telangana	1.75	303.7	6.8 : 2.7 : 1
Uttar Pradesh	7.41	275.9	16.6 : 6.0 : 1
All India	35.0	173.8	6.5 : 2.8 : 1

(Source: FAI, 2021)

Mostly north (189.3 kg ha⁻¹) and south (175.9 kg ha⁻¹) states consume higher fertilizers than east (143.7 kg ha⁻¹) and west (96.2 kg ha⁻¹). In the year 2020-21, the NPK use ratio was widest in North (18:5.8:1) and narrowest in the South (3.9:1.9:1), with intermediate values for West (6.8:3.5:1) and East (3.8:1.7:1). Urea consumption was highest (12.8 million MT) in North zone, while MOP was highest in South (1.33 million MT) as shown in Table 1. Consumption of DAP was highest in the West zone (3.95 million MT), while NPK consumption was highest in South (5.72 million MT). Consumption of fertilizers varied widely across different regions and also in states within the same region. (Table 2)

Table 2 Zone-wise consumption of different fertilizers in India in 2020-21

Zone	Urea (Million MT)	DAP (Million MT)	MOP (Million MT)	NPK (Million MT)	N:P:K ratio
East	4.82	1.49	0.78	1.64	3.8:1.7:1
North	12.8	3.59	0.48	0.71	18.0:5.8:1
South	6.28	1.47	1.33	5.72	3.9:1.9:1
West	11.22	3.95	0.86	4.11	6.8:3.5:1
All India	35.0	10.5	3.44	12.18	6.5:2.8:1

(Source: FAI, 2021)

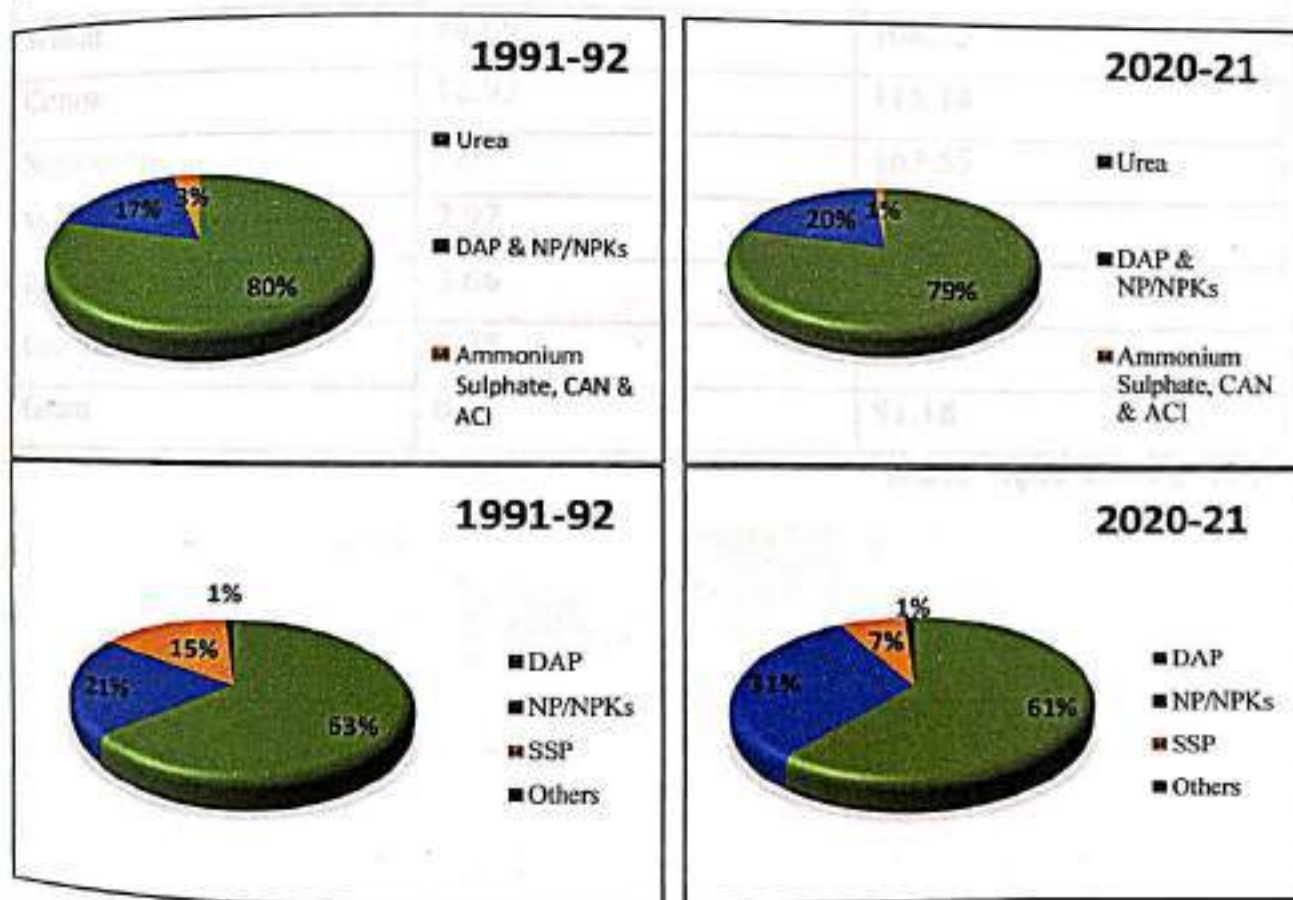


Source: FAI, 2021

4.1.4 Major sources of nitrogen and phosphorus

In India, Urea meets about 80% of all fertilizer nitrogen needs in the country. Data on consumption of fertilizer products reveals that urea continues to be a dominant N source with its share to total consumption being close to 80% during last three decades. (Figure 7). Consumption of N from other sources like DAP, NP, NPK complexes, ammonium sulphate, CAN and ammonium chloride accounted for the rest 20%. Similarly, the share of DAP among P fertilizers continue to be more than 60% even after 30 years. The share of SSP declined by 7% (from 15% in 1991-92 to 8% in 2020-21) and that of NP and NPK complexes registered an increase of 10% (from 21% in 1991-92 to 31% in 2020).

Figure 7 Comparison of the percentage share of nitrogenous and phosphatic fertilizers to nutrient consumption



Source: FAI, 2021

4.1.5 Crop wise N use per hectare

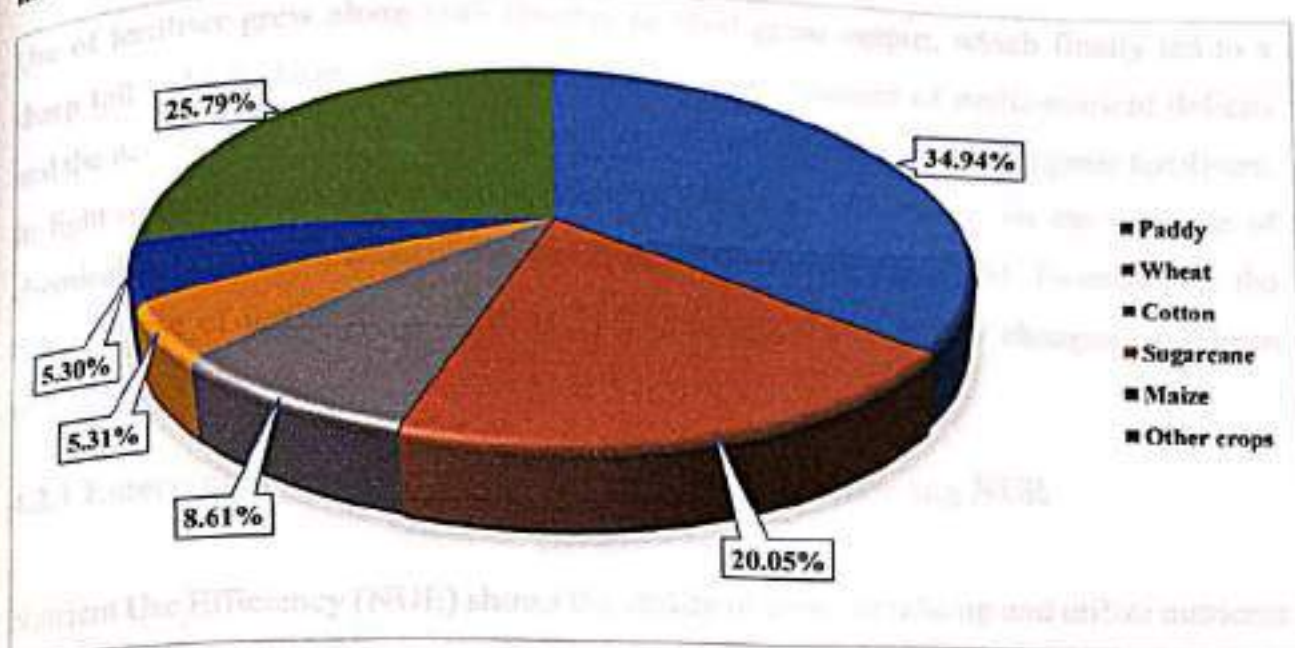
The crop-wise use of nitrogenous fertilizers is highly uneven in India. As per the input survey 2016-17, net N consumption was highest in paddy (52.45 lakh MT) followed by wheat (30.09 lakh MT) and cotton (12.93 lakh MT). Intensity of N use was highest in sugarcane (163.55 kg ha/ha). Among cereals, per hectare N use in India was maximum in wheat (108.75 kg) followed by rice (83.85 kg). (Table 3) As per the input survey 2016-17, paddy and wheat alone consume about 55% of total nutrient N in India. (Figure 8). Fertilizer use, particularly nitrogenous fertilizer use in some crops is more than double the requirement.

Table 3 Crop-wise consumption pattern of N in India in 2016-17

Crop	Total N consumption	N consumption per hectare of gross-cropped area
Paddy	52.45	83.85
Wheat	30.09	108.75
Cotton	12.93	115.14
Sugarcane	7.97	163.55
Maize	7.97	73.31
Rapeseed and Mustard	2.66	71.75
Groundnut	2.38	62.70
Gram	0.41	91.18

Source: Input survey 2016-17

Figure 8 Percent share of different crops in nitrogenous fertilizer consumption in India in 2016-17



Source: Input survey 2016-17

4.2 Impact of technological and policy changes on fertilizer consumption

Use of fertiliser grew along with the rise in food grain output, which finally led to a sharp fall in FUE during the time period. The establishment of multi-nutrient deficits and the decline in soil health were caused by the unbalanced use of inorganic fertilisers. In light of environmental concerns, there has been a lot of focus on the wise use of chemical fertilisers, especially nitrogen fertilisers, all over the world. To encourage the balanced use of fertilizers, a number of technological and policy changes have been made in recent years.

4.2.1 Emerging Technological Interventions for improving NUE

Nutrient Use Efficiency (NUE) shows the ability of crops to take up and utilize nutrients for maximum yields. There are several emerging technological interventions that can help improve NUE in the fertilizer industry. Controlled-release fertilizers are designed to release nutrients slowly over an extended period, reducing the frequent applications and preventing nutrient leaching. Precision farming uses high-tech tools like remote sensing and geographic information systems (GIS) to map and monitor soil fertility and crop growth. This technology helps farmers apply fertilizer more accurately and efficiently, reducing waste and NUE. Nanotechnology involves use of tiny particles to deliver nutrients directly to plant cells. Agricultural machinery plays a major part in application of exact amounts of nutrients and also encourages in Variable Rate of Technology (VRT). By improving the delivery and uptake of nutrients, nanotechnology can enhance NUE and reduce fertilizer waste. New fertilizer formulations, customized fertilizers along with bio-fertilizers contribute to enhancing NUE. (Figure 9)

Figure 9 Emerging technologies for improving NUE



Figure 10 Securing of fertilizer subsidy to India

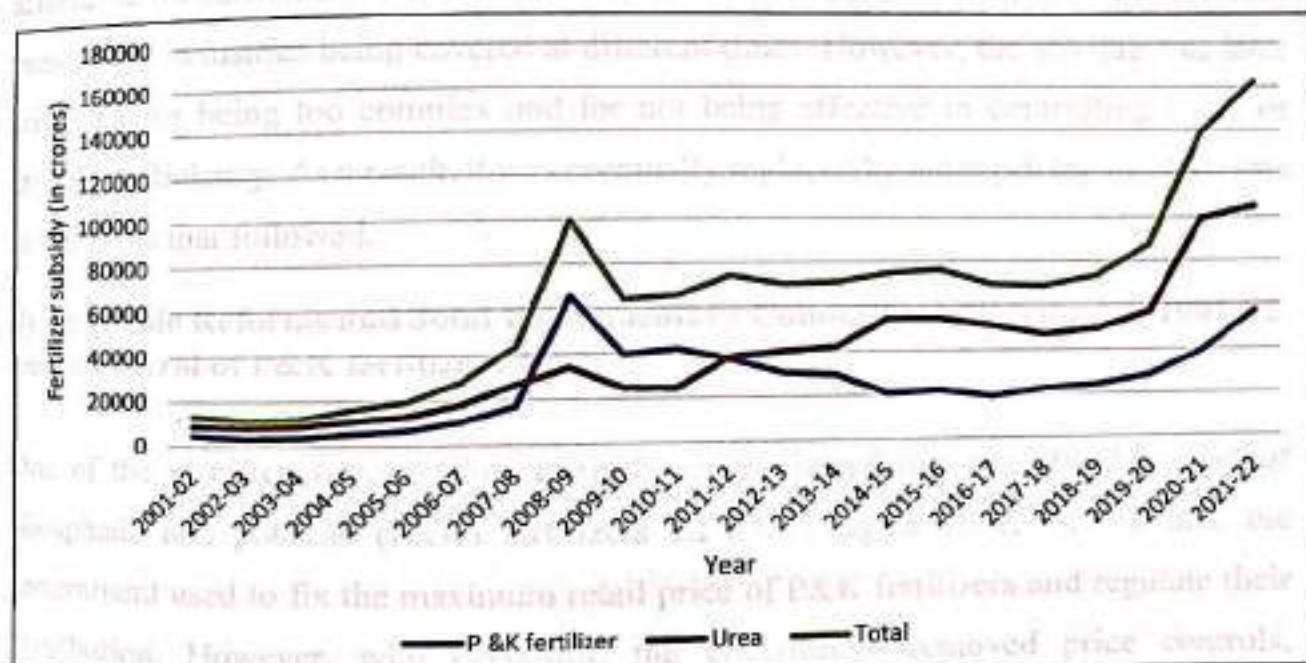
4.2.2 Policy framework for optimum use of fertilizers

In India, fertilizer was brought under Essential Commodities Act 1955 and Government of India (GoI) regulated the sale, price, and quality of fertilizers under Fertiliser (Control) Order (FCO) 1957, which was revised several times till 1985. Revised version is known as Fertiliser (Control) Order 1985. Urea is the only controlled fertilizer, which is sold at the statutory notified uniform sale price. The phosphatic (P) and potassic (K) fertilizers are decontrolled and are sold at indicative maximum retail prices (MRPs).

4.2.3 Scenario of fertilizer subsidy in India

Total subsidy on fertilizers has increased from Rs. 12,907 crore in 2001-02 to Rs. 1,62,184 crore in 2021-22 (Figure 10) and is estimated to be increased to Rs. 1,62,184 crore in 2022-23. For the past 20 years period, subsidy on urea has captured the lion's share in total subsidy except for a period of five years 2007-08 to 2011-12. In 2021-22, almost 65 per cent of total fertilizer subsidy went towards subsidizing urea. The highest share of urea subsidy over the period under consideration was in 2003-04 (almost 72 per cent). In 2002-03, 2014-15 and 2016-17 also, share of urea subsidy in total fertilizer subsidy was more than 70 per cent. After nutrient based subsidy policy was introduced in 2010, share of subsidy on urea in total subsidies started rising again. This increasing subsidy share of urea ultimately led to imbalance in fertilizer nutrient use in the soil. The urea subsidy had increased from 8404 crores in 2001-02 to 104982 crores in 2021-22. The share of urea subsidy among the total subsidy was more than 60 per cent till 2006-07. Though it reduced after that, it has increased again from 2014-15 onwards.

Figure 10 Scenario of fertilizer subsidy in India



Source: Union Budget 2021-22

4.2.4 Important policy changes

Some major historical policy decisions which guided the fertilizer use in the country include:

i) Retention Price Scheme (RPS) (Marathe Committee) 1977

Under the RPS, the price of a product was based on the cost of production plus a fixed percentage of profit. The profit margin was fixed at a level that would provide a reasonable return on investment for the enterprise. The scheme was subject to some efficiency norms, which were intended to encourage enterprises to operate efficiently and reduce costs.

The RPS was seen as a departure from the earlier cost-plus pricing system, which had been criticized for encouraging inefficiency and resulting in higher prices for consumers. The RPS was intended to strike a balance between the need for public sector enterprises to earn a reasonable return on investment and the need to keep prices affordable for consumers. The RPS was implemented in a phased manner, with different sectors and industries being covered at different times. However, the scheme was later criticized for being too complex and for not being effective in controlling costs or ensuring efficiency. As a result, it was eventually replaced by other pricing mechanisms in the years that followed.

ii) Economic Reforms and Joint Parliamentary Committee (JPC) during 1991-92 and decontrol of P&K fertilizers 1992;

iv) National Fertilizer Subsidy (NFS) for P&K fertilizers in 2013

One of the significant steps taken towards economic reforms was the decontrol of phosphatic and potassic (P&K) fertilizers on 25th August 1992. Before this, the government used to fix the maximum retail price of P&K fertilizers and regulate their distribution. However, with decontrol, the government removed price controls, allowing market forces to determine the price of P&K fertilizers. This move aimed to encourage private sector participation in the fertilizer industry and increase competition.

iii) Expenditure Reforms Commission (Geethakrishnan Committee)-2000 and group based new pricing scheme for urea in 2003;

The Expenditure Reforms Commission (ERC) was set up by the Government of India in 2000 under the chairmanship of K.P. Geethakrishnan to suggest measures to improve the efficiency of government spending and reduce wasteful expenditure. The Commission submitted its report in 2001, which contained several recommendations for expenditure reforms in various sectors.

One of the major recommendations of the Geethakrishnan Committee was the introduction of a new pricing scheme for urea. Urea is a widely used fertilizer in India and its prices were controlled by the government, resulting in inefficiencies in the production and distribution of the fertilizer. The Geethakrishnan Committee recommended the adoption of a group-based pricing scheme for urea, which would link the price of urea to its production costs and market demand. The new pricing scheme was implemented in 2003, with the aim of promoting efficiency in the production and distribution of urea and reducing the burden on the government in subsidizing its prices.

Under the group-based pricing scheme, urea production units were classified into four groups based on their location and production costs. Urea prices were then linked to the production costs of the most efficient unit in each group, with a small premium added to cover the cost of production of the other units. This pricing scheme was expected to encourage urea producers to improve their production efficiency and reduce their costs, while ensuring that urea prices remained affordable for farmers.

iv) Nutrient Based Subsidy (NBS)-for P&K fertilizers in 2010

In 2010, to ensure the balanced application of fertilizers, the Government moved towards Nutrient Based Subsidy (NBS) scheme which took the process of decontrolling prices further. The NBS was developed with the goal of introducing some parity between the subsidies paid for nitrogen and for other nutrients, thereby improving the nutrient balance. The scheme, which is in operation till date, covers twenty-one different

kinds of fertilizers, of which DAP, MOP and SSP are the most important. The NBS Scheme introduced three main changes in the system of fertilizer subsidies:

First, the subsidy given to manufacturers was delinked from international prices and the cost of production. Secondly, the subsidy was specified by the national government in terms of nutrient content per unit of nitrogen(N), phosphorus(P), potash(K) and sulphur (S) rather than for different fertilizer products. And thirdly, under the NBS, fertilizer manufacturers were given the freedom to set the retail prices of the fertilizers and the system of government regulation of prices of fertilizers (other than urea) was dismantled. Along with these changes in the system of subsidies, the movement and distribution of twenty per cent of the production and imports of P&K fertilizers were back under the purview of the Essential Commodities Act. It was only for P&K, SSP, and NPK fertilizers. Implementation of the NBS scheme resulted in a sharp rise in the ratio of prices of non-urea to urea fertilizers. In 2018, DAP was 4.5 times more expensive than urea and MOP was 2.5 times more expensive than urea. SSP, which used to be cheaper than urea until 2010 and was the main phosphatic fertilizer used by poor farmers, was 1.4 times more expensive than urea in 2018 (Table 4).

The price of urea has gone up by only 11% between 2000 and 2018, while it has gone up by 150 to 300% for other fertilizers. In addition to heavy subsidy, low urea demand is also a factor.

Table 4 Ratio of retail prices of Urea to DAP, MOP and SSP , 1991-2018

Year	DAP to Urea	MOP to Urea	SSP to Urea
1991	1.50	0.50	0.57
2000	1.93	0.92	0.64
2003	1.93	0.92	0.69
2010	1.76	0.95	0.60
2018	4.53	2.44	1.40

Source: Fertilizer Statistics, 2016-17

The government has fixed the maximum retail price (MRP) of urea at Rs 5,628 per tonne, while the MRPs of other fertilisers are technically decontrolled. The retail price

of urea has remained unchanged during the last decade whereas the price of DAP has increased from 10000 rupees per tonne in 2010-11 to 27000 rupees per tonne in 2021-22. Decontrolling the prices of fertilizers other than urea under the NBS regime has accentuated the urea bias in Indian agriculture and resulted in a sharp decline in the consumption of DAP, SSP and MOP after 2010. Although DAP consumption started to rise slowly after 2014, the fall in MOP consumption has not been arrested. In the financial year 2020-21, It can be inferred from Table 5 that retail price of Urea is lowest in India among major South- Asian countries.

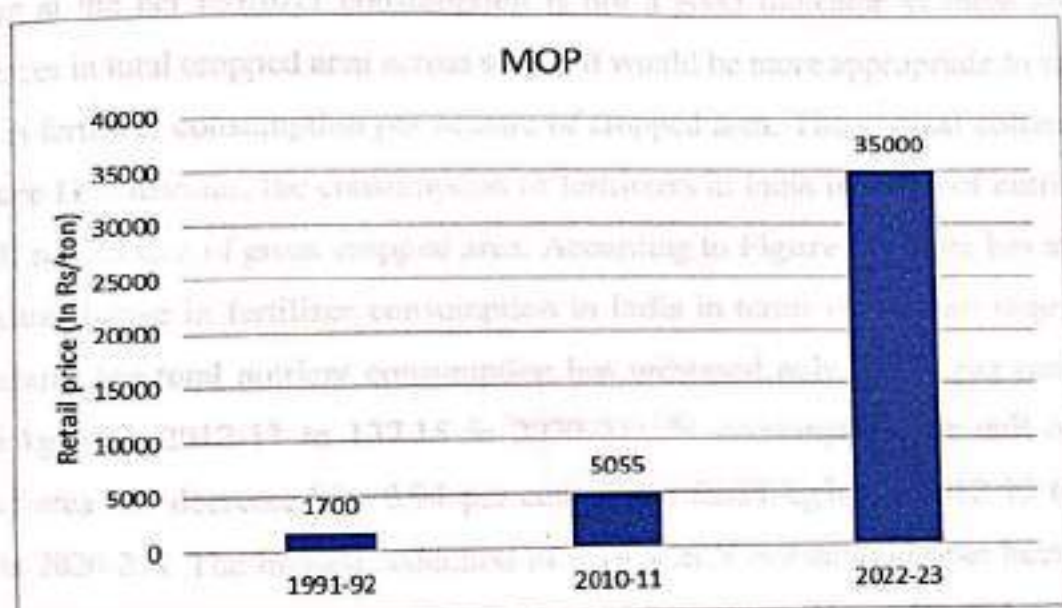
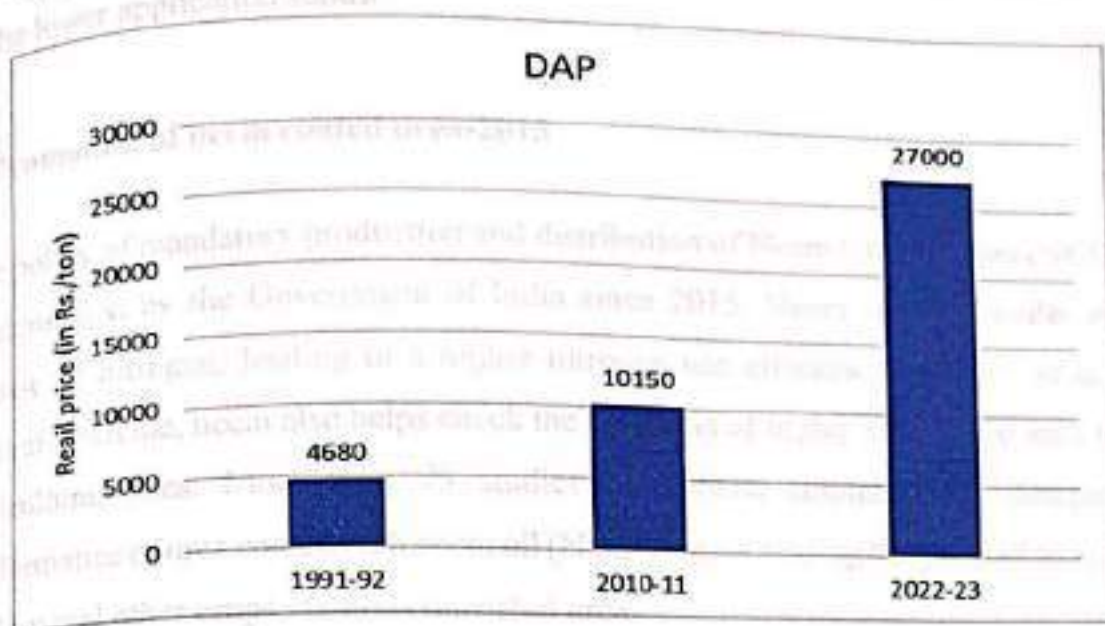
Table 5 Retail prices of Urea in South Asian Countries in 2018

Country	Retail price (\$/ metric ton)
China	352
Pakistan	254
Bangladesh	192
India	80

Source: IFA

The price of urea has gone up by only 11% between 2000 and 2020, while it has gone up by 150 to 300% for other fertilisers. In addition to heavy subsidy, low price inflation is also one reason why farmers prefer Urea. The domestic urea production is uncompetitive; therefore, we are increasingly importing urea. (Figure 11)

Figure 11 Change in retail prices of DAP and MOP from 1991-92 to 2022-23



Source: Fertilizer Statistics

Price control and other government regulations encouraged Urea overuse. Also, they led to the smuggling of fertilizers to neighbouring countries like Bangladesh and Bhutan, where one can sell the fertilizers at a higher price than in India (Economic Survey, 2015-16). The farmers are using more urea at the cost of P&K fertilizers.

Nitrogen application has to be balanced in higher application states and to be increased in the lower application states.

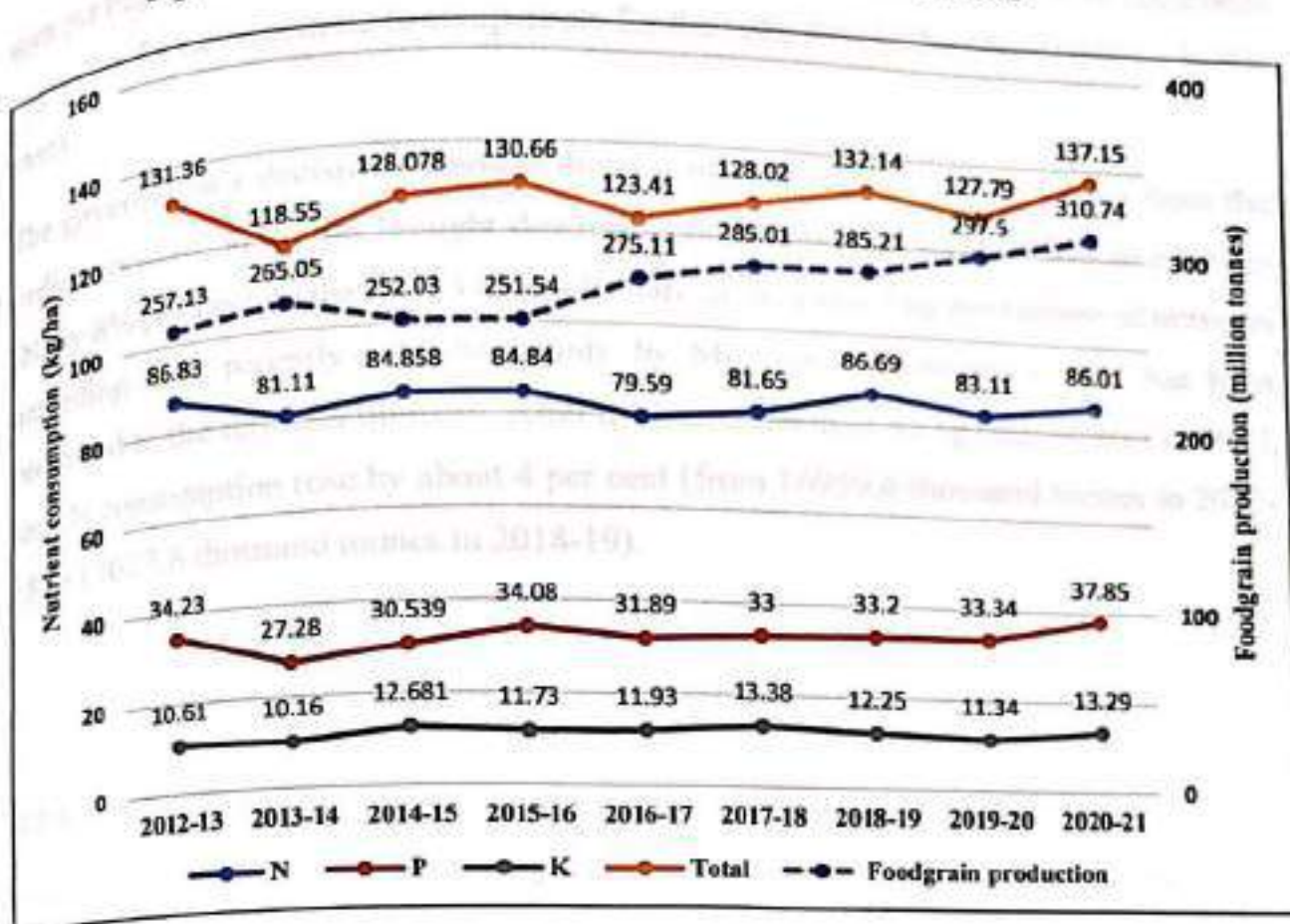
v) Promotion of neem coated urea-2015

The policy of mandatory production and distribution of Neem Coated Urea (NCU) was implemented by the Government of India since 2015. Neem coating results in slow release of nitrogen, leading to a higher nitrogen use efficiency. Besides acting as a natural pesticide, neem also helps check the diversion of highly subsidized urea to non-agricultural uses. More than 75 studies have been conducted to compare the performance of urea coated with neem oil (NOCU) in increasing the yield of rice, wheat, and several other crops vis-à-vis uncoated urea.

Total N consumption fell by about 3.66 per cent (from 17372.3 thousand tonnes in 2015-16 to 16735.9 thousand tonnes in 2016-17).

Looking at the net fertilizer consumption is not a good indicator as there are large differences in total cropped area across states. It would be more appropriate to examine trends in fertilizer consumption per hectare of cropped area. The vertical column chart in **Figure 12** illustrates, the consumption of fertilizers in India in terms of nutrients N, P and K per hectare of gross cropped area. According to **Figure 12**, there has not been significant change in fertilizer consumption in India in terms of nutrient requirement per hectare. The total nutrient consumption has increased only by 4.4 per cent (from 131.36 kg/ha in 2012-13 to 137.15 in 2020-21). N consumption per unit of gross cropped area has decreased by 0.94 per cent (from 86.83 kg/ha in 2012-13 to 86.01 kg/ha in 2020-21). The highest reduction of 6.19% in N consumption per hectare was reported from 2015-16 to 2016-17, the year in which Neem-Coated urea was introduced. There was an increase in food grain production from 251.54 million tonnes to 275.11 million tonnes from 2015-16 to 2016-17. After the introduction of 45 kg bags of urea instead of 50 kg bags in 2017, N consumption has increased by 6.17 per cent from 2017-18 to 2018-19.

Figure 12 Year wise nutrient consumption per unit of GCA



Source: Agricultural Statistics at a Glance (2021)

vi) Reduction in bag size of Urea from 50 kg to 45 kg

With an objective to push farmers to use less urea in proportion to other nutrient fertilizers, such as Phosphorous (P) and Potassium (K), government reduced the weight of one bag of urea from 50 kg to 45 kg. The assumption was that the farmers buy fertilizer based on the number of bags and not by weight.

As per the qualitative assessment conducted by Microsave Consultancy, most of the small and marginal farmers required a smaller number of fertilizer bags, for instance, less than 10 bags. Reducing the weight of the bag resulted in optimizing the use of urea.

Medium and large farmers required more fertilizer bags (for example more than 10 bags). These farmers buy urea by weight in kg rather than by the number of bags. Accordingly, these farmers bought additional bags to compensate for the reduction in

urea per bag. For example, a farmer who would buy 10 bags of urea bought an additional bag under the new scenario to compensate for the reduction in the amount (50 kg in this case).

The government's decision to reduce the size of urea bags to 45 kg in 2017 from the earlier norm of 50 kg has brought down the consumption of the fertilizer or nitrogen (N) by 8% per hectare that is, 15 kg per hectare of urea and 7 kg per hectare of nitrogen according to a recently-published study by Microsave Consultancy that has been presented to the fertilizer ministry. After the introduction of 45 kg bags of urea in 2017, total N consumption rose by about 4 per cent (from 16959.6 thousand tonnes in 2017-18 to 17637.8 thousand tonnes in 2018-19).

4.3.4 Key Findings

The primary survey yielded several key findings that are being used to inform the impact of recent policy changes on use of nitrogenous fertilizers in India. It was found that 38 % of farmers purchased 2 bags of urea each of 50 kg per acre of land (Figure 13). Whereas, 47% of farmers purchased only 1 bag of 50 kg Urea per acre of land. (Figure 16). This shows that most of the farmers purchase fertilizers according to number of bags and not according to quantity of fertilizers.

4.2.5 Primary Survey

In the case of the project "Implications of technological and policy changes on use of Nitrogenous fertilizers in India", a primary telephonic survey was conducted in Bahraich district of Uttar Pradesh and Satna district of Madhya Pradesh.

The primary survey was conducted using a structured questionnaire. The sample was selected using random sampling technique to ensure representativeness and diversity.

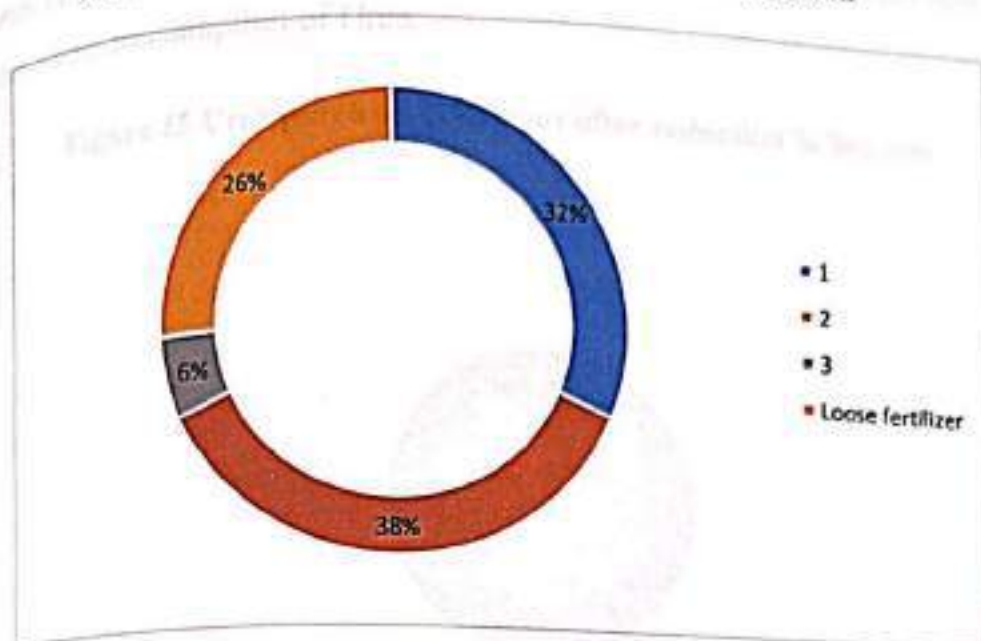
Area	Sample Size
Bahraich, Uttar Pradesh	80
Satna, Madhya Pradesh	20

4.2.6 Key Findings

The primary survey yielded several key findings that can be used to analyse the impact of recent policy changes on use of nitrogenous fertilizers in India.

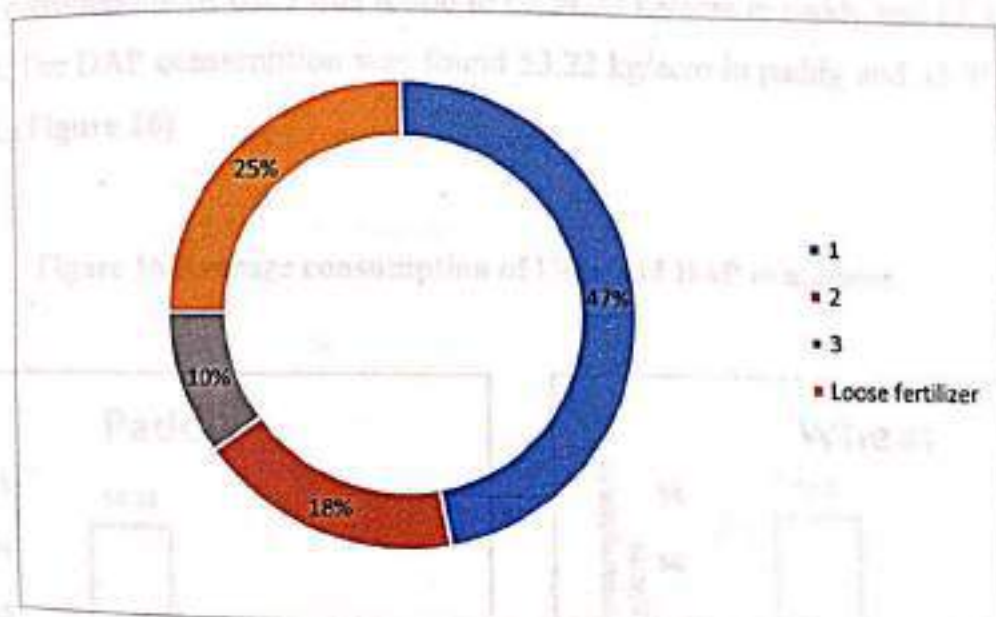
It was found that 38 % of farmers purchased 2 bags of urea each of 45 kg per acre of land (Figure 13). Whereas, 47% of farmers purchased only 1 bag of 50 kg DAP per acre of land. (Figure 14). This shows that most of the farmers purchase fertilizers according to number of bags and not according to quantity of fertilizers.

Figure 13 Urea purchase pattern of farmers in a season



Source: Primary survey conducted in February 2023

Figure 14 DAP purchase pattern of farmers in a season

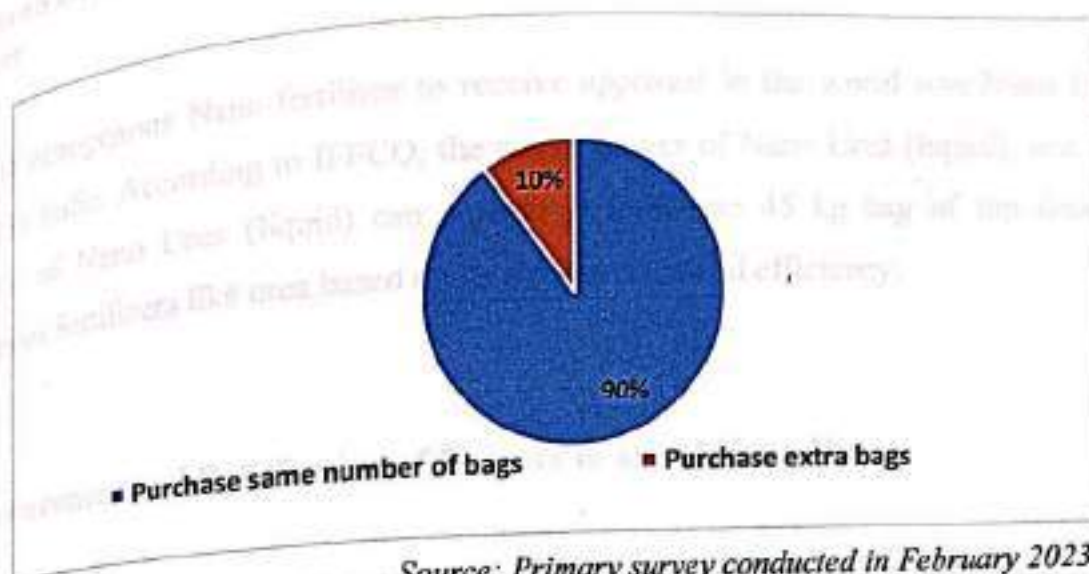


Source: Primary survey conducted in February 2023

It was found that 90% of the farmers continued to use the same number of urea bags per acre of land despite the change in the retail unit of urea bags from 50 kg to 45 kg. Only 10% of the farmers purchased extra bags in order to compensate the reduction in amount

of fertilizer. (Figure 15) Therefore, it can be inferred that reduction in bag size of Urea led to a decrease consumption of Urea.

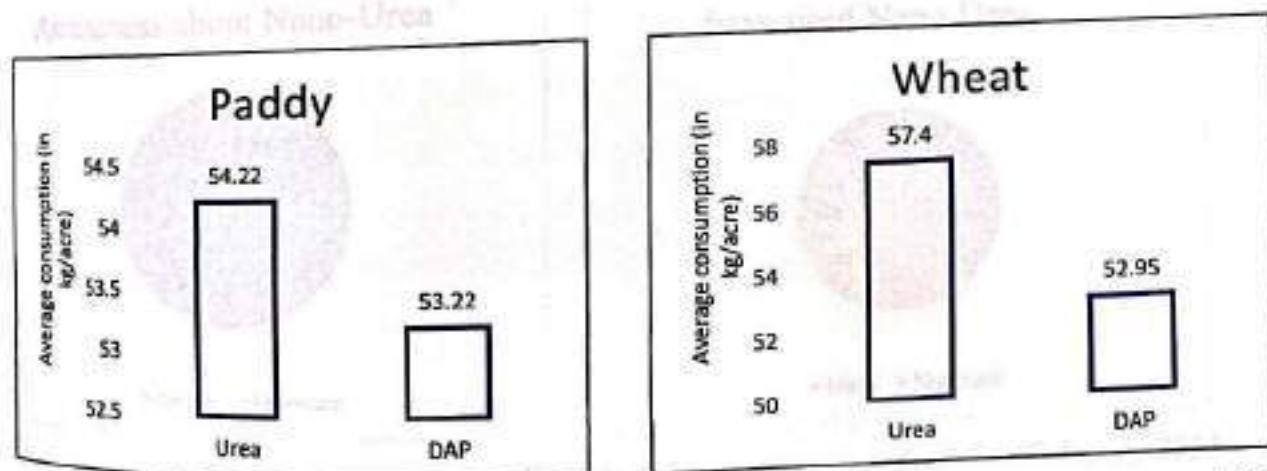
Figure 15 Urea purchase behaviour after reduction in bag size



Source: Primary survey conducted in February 2023

Average consumption of Urea was found to be 54.22 kg/acre in paddy and 57.4 kg/acre in wheat. The DAP consumption was found 53.22 kg/acre in paddy and 52.95 kg/acre in wheat. (Figure 16)

Figure 16 Average consumption of Urea and DAP in a season



Source: Primary survey conducted in February 2023

4.3 Adoption behaviour of Nano urea among selected farmers

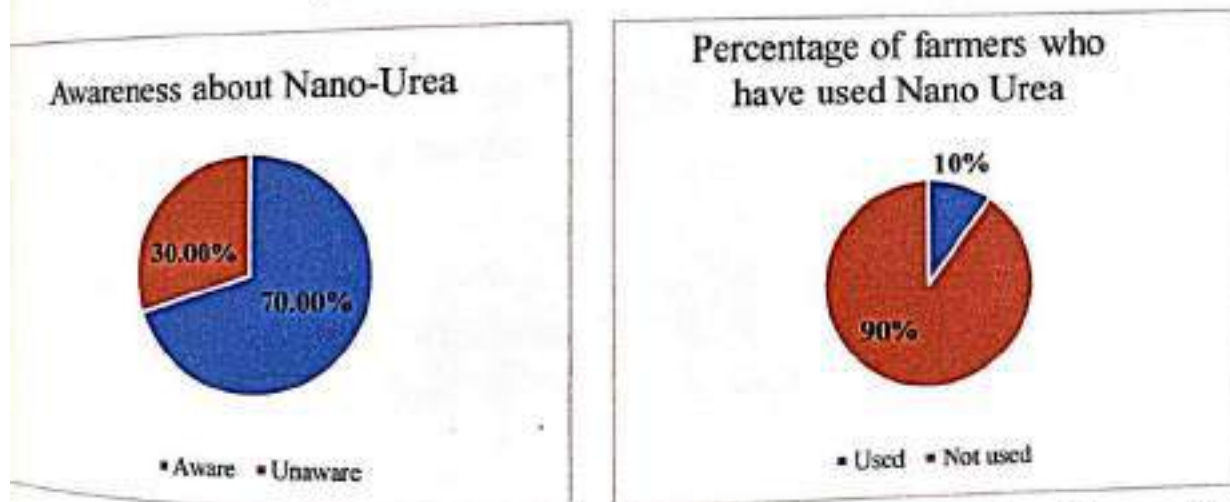
4.3.1 Nano Urea (liquid) – World's 1st Approved Nitrogenous Nano-fertilizer

The first nitrogenous Nano-fertilizer to receive approval in the world was Nano Urea (liquid) in India. According to IFFCO, the manufacturer of Nano Urea (liquid), one 500 ml bottle of Nano Urea (liquid) can replace at least one 45 kg bag of top dressed nitrogenous fertilizers like urea based on its performance and efficiency.

4.3.2 Awareness and Satisfaction of farmers in use of Nano Urea

Awareness studies play crucial role in enriching understanding about the awareness-decision paradoxes and their determinants (Paul et al. 2017). A better awareness is an indication of progressiveness (Kumar et al. 2015). Results of the present study show that about 70% farmers were aware about the introduction of Nano urea. But, only 10% of the farmers actually used Nano urea for growing crops. (Figure 17)

Figure 17 Awareness and Use of Nano Urea



Source: Primary survey conducted in February 2023

On an average, after the adoption of Nano Urea, farmers applied only half of the amount of normal urea than they used to apply without Nano urea. Almost all the farmers used Knapsack spray pump for spraying Nano urea. The farmers who applied Nano Urea were satisfied with the results obtained from its use in terms of increase in yield and reduction in input cost.

The present study suggests that awareness of farmers regarding Nano Urea is quite encouraging. Farmers when asked possible reasons for not using Nano urea cited reasons like unawareness about benefits of Nano-urea, their wish to continue with a product that is in use since years rather than going for a newly launched product and lesser availability of Nano urea.

CHAPTER-5 CONCLUSION

In conclusion, balanced use of fertilizers is a critical issue that requires immediate attention from government and other stakeholders. The consumption of nitrogenous fertilizers is on the rise after the green revolution. The government has taken a number of steps in recent years to encourage balanced use of fertilizers. But these changes have proved to be insufficient to bring about balanced use of fertilizers. Urea's cheap price has encouraged the usage of nitrogen fertilizer. On the other hand, fixed subsidy on P & K fertilizers has been brought down gradually leading to high retail prices of these products. This has led to significant distortion in prices of fertilizer products and thereby NPK use ratio. The development and use of complex fertilisers other than DAP, which is comparably more expensive, should be encouraged through price and subsidy programmes. There is a need for creating balance between prices of urea and other fertilizers. A gradual increase in urea price in small increments is warranted. Subsidy on P & K fertilizers under NBS policy should also be suitably adjusted. The objective of policy correction should be to restore the price ratio of DAP: urea close to 2:1 and MOP: urea close to 1:1. There is a need for promotion of more innovative fertilizer products like Nano urea and other liquid fertilizers. Although the awareness level of Nano urea in case of farmers is significant, its adoption ratio is not encouraging. More emphasis must be given on dissemination of knowledge and information about the benefits of using Nano Urea at recommended doses through awareness campaigns, training, capacity building programmes, farm literature, mass media and other modes of technology transfer.

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